

A Conditional Boolean Interface

Optional Supplement to the Chain-Ring Resource Classification

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This note records an optional symbolic construction. The main article does not depend on it. It supplies neither a physical probability model nor a preparation theory closed under all conditioning operations.

1 Scalar enrichment and support

Let B be the Boolean ring of formulas modulo logical equivalence and let $A = \mathbb{F}_2[u]/(u^4)$. Every element of B is idempotent, whereas the only idempotents in a commutative local ring are 0 and 1: if $e(1 - e) = 0$, one of e and $1 - e$ is a unit. A unital map $B \rightarrow A$ therefore cannot generate u .

Instead choose the scalar extension $R = B \otimes_{\mathbb{F}_2} A$. Each element of R is uniquely $\sum_{j=0}^3 f_j u^j$, with $f_j \in B$. A Boolean valuation $v : B \rightarrow \mathbb{F}_2$ extends coefficientwise to $v_A : R \rightarrow A$. For a chosen state $\psi \in R^n$ and covector $e \in (R^n)^*$, write $e\psi = \sum_j g_j u^j$ and define

$$\text{Supp}(e\psi) = g_0 \vee g_1 \vee g_2 \vee g_3.$$

Then uniqueness of the polynomial expansion gives

$$v(\text{Supp}(e\psi)) = 1 \iff v_A(e)v_A(\psi) \neq 0.$$

Disjunction is essential; nonzero testing is neither additive nor multiplicative. The construction is conditional on the chosen enrichment, state, effects and tensor base. It is not a unique interpretation of bare Boolean formulas and does not automatically extend to mixed-characteristic scalar rings.

2 Coordinate projectors over the symbolic ring

Let $B_0 \in \text{GL}_n(R)$ and let P_i be the standard coordinate projections. Set $J_i = B_0^{-1}P_iB_0$. Direct multiplication gives

$$J_i^2 = J_i, \quad J_i J_j = 0 \ (i \neq j), \quad \sum_i J_i = I.$$

For every $\psi \in R^n$,

$$B_0 J_i \psi = P_i B_0 \psi.$$

Since B_0 is invertible, $J_i \psi = 0$ if and only if $P_i B_0 \psi = 0$, which is equivalent to $(B_0 \psi)_i = 0$. This proof is valid over any commutative coefficient ring. It uses no division by the measured amplitude and no unit-coordinate assumption on individual columns. Applying v_A to the inverse identity preserves invertibility and the projector identities, so the same statement holds after every valuation.

The unit-coordinate shortcut valid over a local ring cannot be used over R , which need not be local. For a nontrivial idempotent $t \in B$, the symbolic matrix

$$C = \begin{pmatrix} t & 1+t \\ 1+t & t \end{pmatrix} \quad \text{satisfies} \quad C^2 = I,$$

but neither entry in either column is a unit: t and $1+t$ are nonzero orthogonal idempotents. The inverse-identity argument above avoids this obstruction.

These projectors supply repeatable linear branches, not probability normalization. They do not repair primitive-state closure: over A , conditioning the primitive resource $\text{diag}(1, u)$ on its second standard row gives the nonprimitive nonzero vector $(0, u)$.